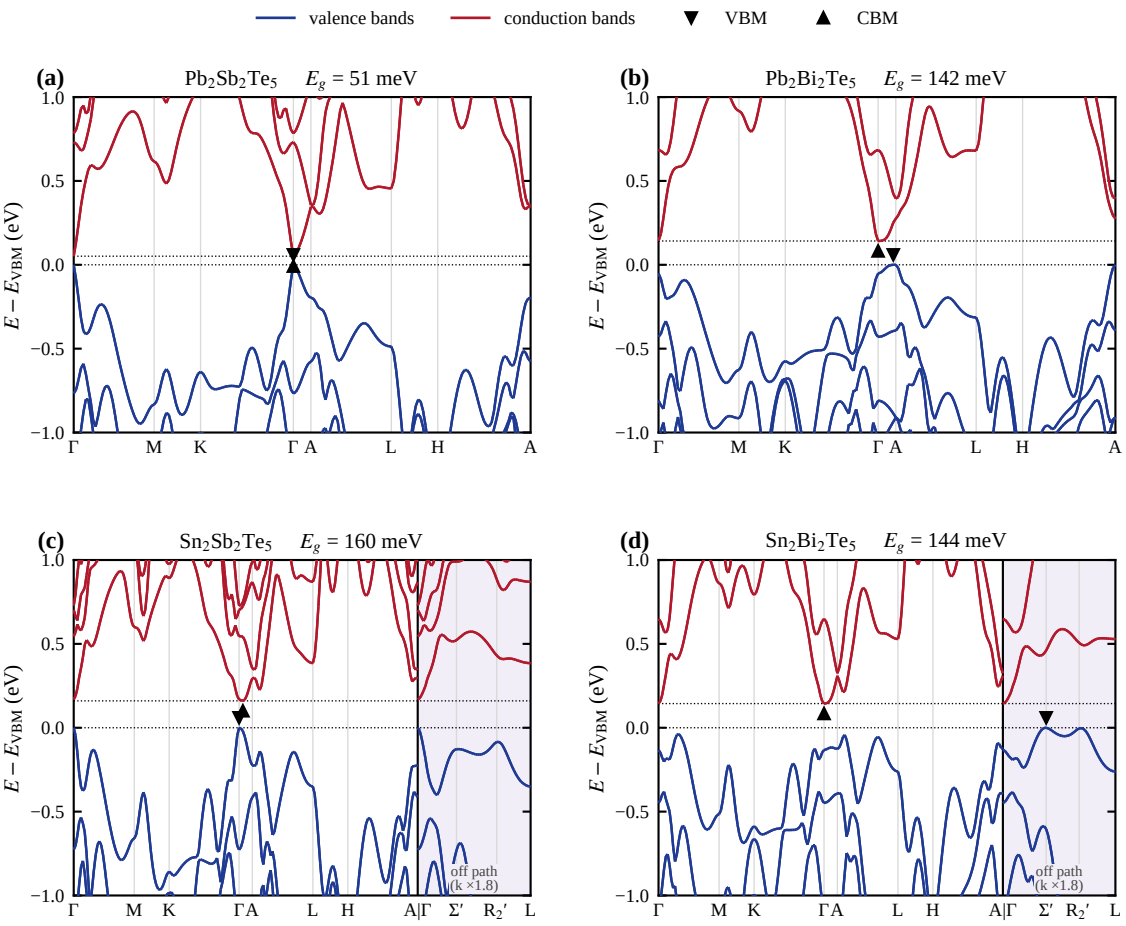

valence bands
conduction bands
VBM
CBM
(a)
Pb2Sb2Te5 Eg = 51 meV
(b)
Pb2Bi2Te5 Eg = 142 meV
(c)
Sn2Sb2Te5 Eg = 160 meV
(d)
Sn2Bi2Te5 Eg = 144 meV
E − EVBM (eV)
1.0
0.5
0.0
−0.5
−1.0
Γ M K Γ A L H A
Γ M K Γ A L H A|Γ Σ′ R2′ L
off path
(k ×1.8)